



Review

Advances in investigating microcystin-induced liver toxicity and underlying mechanisms

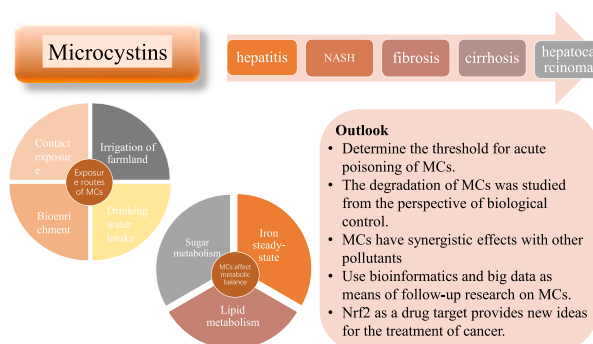
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HIGHLIGHTS

- The toxicity of microcystin to aquatic and terrestrial animals was compared.
- The microcystin-induced hepatic metabolic imbalance was discussed.
- Provide new ideas and methods for developing microcystin-targeted drugs
- The study of microcystins is prospected from the perspective of bioinformatics.

GRAPHICAL ABSTRACT



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ABSTRACT

Microcystins (MCs) are a class of biologically active cyclic heptapeptide pollutants produced by the freshwater alga *Microcystis aeruginosa*. With increased environmental pollution, MCs have become a popular research topic. In recent years, the hepatotoxicity of MCs and associated effects and mechanisms have been studied extensively. Current epidemiological data indicate that long-term human exposure to MCs can lead to severe liver toxicity, acute toxicity, and death. In addition, current toxicological studies on the liver, a vital target organ of MCs, indicate that MC contamination is associated with the development of liver cancer, nonalcoholic fatty liver, and liver fibrosis. MCs produce hepatotoxicity that affects the metabolic homeostasis of the liver, induces apoptosis, and acts as a pro-cancer factor, leading to liver lesions. MCs mainly mediate the activation of signaling pathways,

Abbreviations: MC, microcystins; EROD, ethoxyresorufin-O-deethylase; ERND, ethoxyresorufin-N-deethylase; ANH, aniline hydroxylase; UA, uric acid; BUN, blood urea ammonia; SCr, creatinine; HSC, hepatic stellate cell; OATP, organic anionic protein transport peptide; BBB, blood-brain barrier; BCFB, blood-cerebrospinal fluid barrier; NASH, non-alcoholic steatohepatitis; HCC, hepatocellular carcinoma; ALT, alanine aminotransferase; ALP, alkaline phosphatase; NAFLD, nonalcoholic fatty liver disease; OXPHOS, oxidative phosphorylation; PP1, protein phosphatase-1; PP2A, protein phosphatase-2.

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such as the ERK/JNK/p38 MAPK and IL-6-STAT3 pathways, which leads to oxidative damage and even carcinogenesis. Moreover, MCs can act synergistically with other pollutants to produce combined toxicity. However, few systematic reviews have been performed on these new findings. This review systematically summarizes the toxic effects and mechanisms of MCs on the liver and discusses the combined liver toxicity effects of MCs and other pollutants to provide reference for subsequent research. The toxicity of different MC isomers deserves further study. The detection methods and limit standards of MCs in agricultural and aquatic products will represent important research directions in the future. Standard protocols for fish sampling during harmful algal blooms or to evaluate the degree of MC toxicity in nature are lacking. In future, bioinformatics can be applied to offer insights into MC toxicology research and potential drug development for MC poisoning. Further research is essential to understand the molecular mechanisms of liver function damage in combined-exposure toxicology studies to establish treatment for MC-induced liver damage.

1. Introduction

Microcystins (MCs) are the most common cyanobacterial toxins in eutrophic waters and pose a severe threat to aquatic life and human health worldwide (Shi et al., 2021). Massive blooms and surface scums of cyanobacteria regularly emerge in the Baltic Sea (Löptien and Dietze, 2022). MCs are considered the most widely distributed cyanobacterial toxins in nature, and they are present at high levels (up to 1 % of dry weight) in cyanobacterial biomass (Bláha et al., 2009).

The World Health Organization (WHO) derived a tolerable daily intake (TDI) value of MC-LR for human health risk assessment. Large-scale cyanobacterial blooms have occurred in >100 countries and pose a serious global public health risk (Harke et al., 2016). As cyanobacteria proliferate, the ecological balance of the hydrosphere is disrupted because of the release of MCs into the water body (Díez-Quijada et al., 2019). Toxic cyanobacterial blooms occur seasonally in the James River estuary during the dry summer months, and the highest chlorophyll and cyanobacterial levels are observed in the tidal freshwater segment (Bukaveckas et al., 2018). MCs were detected in the viscera of fish, gastropods, shrimps, and clams in the Mekong River basin of Vietnam at levels as high as 11,039 µg/L (Trung et al., 2018). In water contaminated with MCs, MC concentrations typically remain below 100 µg/L, although environmental conditions (nutrients, pH, and water temperature) may increase the concentrations above 1000 µg/L. MCs accumulate gradually in the soil and then are released and migrate to the subsoil with precipitation events. When MCs reach deeper soil layers, they can seep into groundwater and affect the quality of drinking water (Melaram et al., 2022). A bloom event in Chaohu Lake in China caused the MCs concentration in groundwater in the surrounding area to reach 1.07 µg/L, which posed a serious threat to drinking water sources (Yang et al., 2016). In addition, MC contamination was observed in all four public water systems in Zaria and Nigeria (Chia and Kwaghe, 2015). Currently, water pollution from MCs has spread to lakes and oceans worldwide, and a variety of isomers has been observed (Table 1). MCs have several isomers such as MC-LR, MC-RR, and MC-YR owing to the different compositions of the two variable amino acids in its polypeptides. The production of MCs and their isomers is closely associated with epidemics and environmental problems (Zhu et al., 2016).

MCs have stable chemical properties, and traditional water treatment methods cannot easily remove MCs; currently, biodegradation is the most successful solution for removing MCs from the natural environment (Sheik Abdul and Marnewick, 2020). Water pollution by MCs also poses a severe threat to farmland ecosystems. Irrigating fields with MC-contaminated water leads to the accumulation of toxins in agricultural products (e.g., bok choy and tomatoes) (Zhi-ming et al., 2011). Furthermore, the spread of MCs up the food chain endangers human and animal health. Cyanobacteria are also frequently fished and deposited in farmland soil as organic fertilizers, which leads to the contamination of agricultural land soil (Xiao-jing et al., 2015). Irrigation with MC-contaminated water leads to the accumulation of toxins in agricultural products, which are then ingested by humans and animals (Zhe-shen et al., 2016).

Consequent to their environmental effect, MCs pose a severe threat

to human health. When water bodies are contaminated, the primary human exposure risk is via the consumption of MC-contaminated organisms, including fish, shrimp, and shellfish in particular (Fig. 1) (Humbert, 2010). Fishermen who regularly operate in freshwater

Table 1
Contamination characteristics of some typical MC-contaminated water bodies.

Country region	Lakes/oceans	MC type	MC concentration	References
Zaria, Nigeria	Public water supply reservoir	MCs	3.90 ± 0.06 µg/L	(Chia and Kwaghe, 2015)
Lucas County, Ohio	Collins Park Water Treatment Plant	MCs	3.19 µg/L	(McCarty et al., 2016)
China	Dianchi	MCs	23.06 µg/L	(Wei et al., 2022)
China	Tai Lake	MCs	1.00 µg/L	(Wei et al., 2022)
Sancul dara	Freshwater pool	MCs	2.19–39.60 g/L	(Singh et al., 2015)
Lakshmikund	Freshwater pool	MCs	15.22–128.14 µg/L	(Singh et al., 2015)
China	Tai Lake Gonghu Bay	MC-LR	17 µg/L	(Wang et al., 2010b)
Jilin, China	Artificial wetland	MC-LR	2.411 µg/L	(Yue, 2021)
Hangzhou, China	Paste Sand River	MC-LR	31.6–472 ng/L	(Ming et al., 2016)
Shanghai, China	Dianshan Lake	MC-LR	(13.98 ± 16.99) ng/L	(Xi et al., 2018)
Shaoxing, China	Xiaoshao River Network, Cao'e River, Puyang River	MC-LR	0–4.28 µg/L	(Sai et al., 2018)
Anhui, China	Chaohu Lake	MC-LR	1.07 µg/L ⁻¹	(Yang et al., 2016)
Three Gorges, China	Three Gorges Reservoir Area	MC-LR	0.57 µg/L ⁻¹	(Xu et al., 2011)
Athens, Greece	Marathon Lake	MC-YR, MC-LR, MC-RR,	717 ng/L ⁻¹ , 451 ng/L ⁻¹ , 174 ng/L ⁻¹	(Kaloudis et al., 2013)
France	Brittany Coast	MC-YR, MC-LR, MC-RR,	544 ng/g fresh weight	(Lance et al., 2021)
China	Huai River Basin	MC-LF, MC-LR, MC-RR, MC-YR, MC-LW	17.731 µg/L ⁻¹	(Tian et al., 2011)
Algeria	Lake of the Ozarks	MC-RR, MC-LR, MC-FR, MC-WR, MC-YR, MC-LA, MC-(H4) YR, MC-HiLR	/	(Bouhaddada et al., 2016)

systems have much higher blood MC levels than those recommended by the WHO (Melaram et al., 2022). MCs can bioaccumulate in fish, crabs, lobsters, abalones, and filter-feeding bivalves (shellfish), such as mussels, oysters, scallops, and clams, leading to the poisoning of seafood consumers (Shi et al., 2021). Further, ingesting food and dietary supplements contaminated with cyanobacteria, drinking water contaminated with cyanogen toxins, or engaging in recreational activities in water containing MCs can lead to human exposure to MCs in large quantities.

Approximately 60,000 cases of human poisoning occur worldwide each year, with an overall mortality rate of approximately 1.5 % (Ajani et al., 2017). MCs have multiple toxic effects, including genotoxicity, neurotoxicity, immunotoxicity, and potential carcinogenic properties (AlKahtane et al., 2020; Li et al., 2019; Xu et al., 2020). Studies have shown that MCs can directly induce cardiovascular disease (Cao et al., 2019) and kidney disease based on abnormal blood uric acid (UA), blood urea ammonia (BUN), and creatinine (SCr) (Xu et al., 2020).

Numerous studies have confirmed the potential link between MC exposure and liver disease. The toxicity of MCs exhibits significant organ selectivity, and the liver is a vital target and effector organ. Exposure to MCs at different doses and durations can cause varying degrees of damage to the liver. Acute exposure to MCs may lead to fatty liver disease and liver failure, (He et al., 2017a), whereas long-term exposure to MCs can lead to chronic liver injury and inflammation (Lin et al., 2018). Moreover, a mouse liver model was used to confirm that liver fibrosis occurred in mice exposed to MC-LR at 15 µg/kg and 30 µg/kg for 14 d (Gu et al., 2020).

The damage caused by MCs to the human liver is directly manifested as liver enlargement, congestion, increased liver body proportion, liver cell destruction, liver hemorrhage, and liver necrosis. Mice exposed to MCs exhibited significant hepatophilia. ATP-dependent transporters are required for the transmembrane transport of MCs, whereas MC toxicity is primarily restricted to the liver via organic anion transporters on the cell membrane (Zhi-jian et al., 2002). In mice exposed to MCs, some hepatocyte cords were damaged and the liver sinuses were dilated (Lijiang et al., 2013). The mitochondrial function of hepatocytes was also

altered by exposure to MCs (He et al., 2021), and oral MCs induced apoptosis in mice and stress in the endoplasmic reticulum (Huang et al., 2011). The slow staining of liver cells by low-dose MCs not only increased reactive oxygen species (ROS) levels but also damaged the DNA and mitochondrial functions of liver cells, significantly decreasing the activity of ethoxyresorufin-O-deethylase (EROD) and ethoxyresorufin-N-deethylase (ERND) and increasing the activity of aniline hydroxylase (ANH) in the mouse liver. Damage to the function of liver cytochrome P450 was also observed (Zhang et al., 2015). MCs cause hepatotoxicity as well as renal toxicity, neurotoxicity, and genetic toxicity. However, the liver is the most crucial target organ and hepatotoxicity remains one of the most critical manifestations of MCs. MCs usually induce liver fibrosis, nonalcoholic fatty liver disease, liver cancer, and other serious diseases, thus posing a severe threat to the health of exposed organisms. Moreover, an epidemiological investigation confirmed the significant correlation between serum MC content and liver cancer (Krishnan et al., 2020).

Herein, we review the toxicological mechanisms of MCs based on hepatotoxicity cases published over the past ten years. The focus of this review is on liver toxicity caused by MCs from the aspects of liver metabolic environment homeostasis, programmed necrosis, liver fibrosis, and liver inflammation, and the objective is to summarize the mechanisms underlying the ability of MCs to induce liver disease and relevant evidence for its liver toxicity, point out the research gaps in this field, and discuss future research directions. The findings will provide a reference for subsequent clinical diagnosis and treatment of MC poisoning.

2. Liver toxicity of MCs to animals in different habitats

2.1. Liver toxicity of MCs to terrestrial animals

MCs can cause acute poisoning in common terrestrial animals, such as sheep, rabbits, mice, guinea pigs, chickens, and ducks, where MC ingestion causes liver congestion and focal necrosis. Most animal poisoning-related deaths caused by MCs are caused by excessive MC

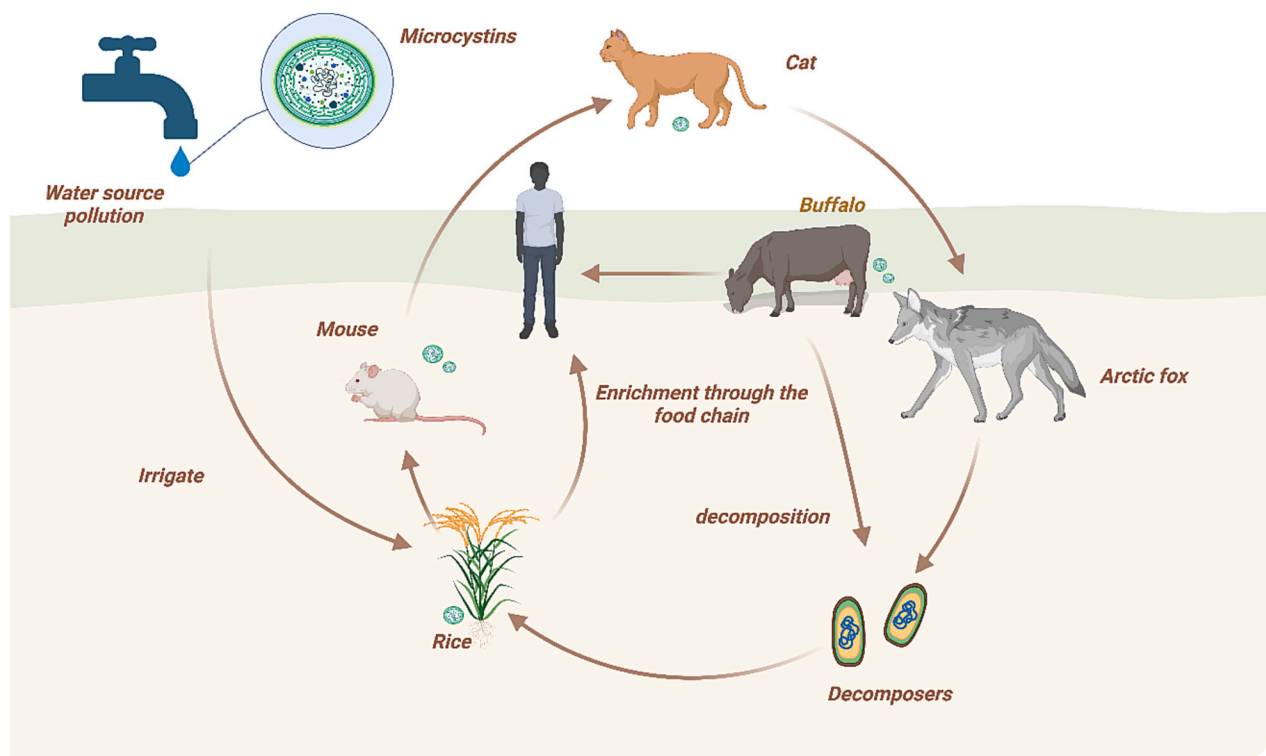


Fig. 1. Exposure pathways of microcystins in humans and terrestrial animals.

intake. It has been found that dogs may be attracted to the smell of cyanobacteria and might swallow substantial amounts of floating mats accumulated on leese shorelines (Codd et al., 1992). Consistent with these findings, three dogs died after ingesting *Phormidium* mat material that washed ashore along Lake IJmeer in the Netherlands in the spring of 2011, and cyanophycin poisoning appeared to be the cause of death (Faassen et al., 2012). In the same year, two dogs died of poisoning after walking along the shores of Lake Amsternere in the Netherlands, where cyanobacterial scum had washed ashore. After eating the cyanobacteria, the dogs experienced gastrointestinal bleeding, loss of appetite, slow movement, and then death (Lüring and Faassen, 2013). In terrestrial animals, MC poisoning occurs through direct ingestion of MCs or drinking water from water sources after blooms. In Central Michigan, three cows died after drinking water contaminated by blooms, and autopsies revealed diffuse necrosis of the liver and kidneys (Fitzgerald and Poppenga, 2016). Studies have shown that approximately 50%–70% of ingested MCs are distributed in the liver (McLellan and Manderville, 2017). MCs enter animal cells in various ways, although the primary route is through the transmembrane organic anionic protein transport peptide (OATP) (Stieger and Hagenbuch, 2014). Studies have shown that OATP1B1, OATP1B2, OATP1B3, and OATP1A2 are involved in the cellular uptake of MCs. After MC exposure, these peptides become imbalanced in the liver tissues, thus affecting the balance of the body's transmembrane proteins (Menezes et al., 2013). OATP1B1 and OATP1B3 are bile acid transporters located on the basement membrane of hepatocytes that are responsible for transporting bile acids from the hepatic sinuses to liver cells. MCs can be transported across the blood-brain barrier (BBB)/blood-cerebrospinal fluid barrier (BCFB) in an OATP/OATP-dependent manner and subsequently induce cytotoxicity by inhibiting protein phosphatase (PP) (Feurstein et al., 2009).

Previous investigations have shown that MCs accumulate in the livers of terrestrial animals (small brown bats and deer) and some domestic animals (e.g., cattle and pigs). (Handeland and Østensvik, 2010; Redouane et al., 2023; Russell et al., 2015). Pets and many aquatic creatures experience skin irritation, vomiting, liver cancer, and death. Some of these animals are nationally protected (e.g., the North American bat) and face serious conservation threats (Frank et al., 2014). However, limited data are available on their exposure to MCs. To protect these animals from MC toxicity and ensure that their habitat is not threatened, we believe that it is very important to study the hepatotoxicity of MCs in terrestrial animals. Soil fauna play an essential role in ecosystems. On one hand, soil fauna actively assimilate various useful substances to build themselves; on the other hand, they return their excrement to the environment and constantly transform it. The toxic effects of MCs on soil fauna still need to be studied, including the effects of acute (such as survival and feeding) and chronic toxicity (such as growth and reproduction), changes in physiological and biochemical indicators, such as the hepatopancreas (such as protease and phosphatase activities), and effects on behavior and distribution.

2.2. Liver toxicity of MCs to aquatic animals

Large cyanobacterial outbreaks associated with toxins, including MCs, have been speculated to cause diseases and disorders in aquatic animals, including fish (Shahmohamadloo et al., 2023b). At sufficient concentrations, MCs can disrupt the growth and development of fish embryos, affect fish behavior, and accumulate in the liver and edible muscle tissues; thus, MCs represent an emerging threat to fish populations (Malbrouck and Kestemont, 2006; Shahmohamadloo et al., 2021; Shahmohamadloo et al., 2022). A number of studies have shown that before and after bloom events, MCs accumulate in fish tissues, especially in the liver and muscles, and are found across diverse fish species (Amrani et al., 2014; Chang-gao et al., 2022; Deblois et al., 2008; Ni-na et al., 2011; Shahmohamadloo et al., 2023a; Shahmohamadloo et al., 2023c).

The toxic effects of MCs on fish livers have been studied extensively

under different exposure pathways, including their effects on the hatching, development, behavior, and growth of fish embryos. A previous study raised Kras^{V12}, a transgenic zebrafish that can induce a liver tumor, in 3 µg/L MC-LR exposure solution, and tumors appeared in the liver of female Kras^{V12}. The results showed that MC-LR acted via the Wnt signaling pathway. Decreased protein phosphatase 2A (PP2A) and PP2A subunit transcription levels lead to tumor development (Zi-jing, 2022). The activities of superoxide dismutase (SOD) and catalase (CAT) in the hepatopancreas of juvenile grass carp were significantly increased after intraperitoneal injection of 100 µg/kg⁻¹ MC-LR, indicating that MC-LR has a significant stress effect on the hepatopancreas antioxidant system of juvenile grass carp (Li-li et al., 2019). After 500 µg/L MC-LR induction in vitro for 24 h, single-cell microgel electrophoresis showed that the treatment group presenting significant comet tailing compared with the control group (no MC-LR treatment group), indicating that MC-LR could cause DNA damage in Nile tilapia hepatocytes (Xiu-xia, 2009). Studies have shown that the level of inducible GSTR genotypes in the livers of freshwater fish with different ecological habits may be related to the detoxification ability of the liver (Di et al., 2020). Such findings can provide an essential theoretical basis for preventing and controlling cyanobacterial blooms and have great practical significance. Balneation of rainbow trout (*Oncorhynchus mykiss*) and lake trout (*Salvelinus namaycush*) with sublethal concentrations of MC-LR for brief periods (24 h) also demonstrated that such exposure can lead to accumulation of MCs in the liver and the dysregulation of proteins related to oxidative stress and cancer formation; moreover, these effects were sustained even after the fish were reintroduced to clean water (Shahmohamadloo et al., 2021; Shahmohamadloo et al., 2022).

The main difference in MC exposure pathways between terrestrial and aquatic animals is that terrestrial animals are usually poisoned by accidental contact or brief accidental ingestion of MC-containing organisms while aquatic organisms accumulate toxins in their bodies over long periods of time because they reside in MC-containing waters. However, aquatic organisms show lower mortality rates than terrestrial animals, which may be related to various factors in aquatic organisms that confer resistance to MCs. The mortality rate of aquatic organisms exposed to MCs over a long period perhaps lower than that of exposed terrestrial animals, which can be used as a reference for the development of targeted drugs for the treatment of MC poisoning in later stages. Although MC exposure varies between terrestrial and aquatic organisms, the toxic effects are similar. Analytical standards for precisely determining the MC content in aquatic organisms have been developed to better assist with environmental analyses and human health risk assessments associated with fish consumption (Anaraki et al., 2020). Considering that MC toxicity events may be increasing due to various climatic factors (Hellweger et al., 2022), diagnostic criteria must be established for MC poisoning events. A number of analytical standards are available to quantify cyanobacterial toxins in water samples; however, standard protocols should be established for fish sampling during harmful algal blooms. The detection methods and limit standards for MCs in agricultural and aquatic products represent important research directions in the future (Studer et al., 2022).

3. Toxicological study of MCs in the human liver

When MCs enter liver cells, they cause metabolic disorders in the liver, alter the cell morphology, and induce apoptosis. As a large number of MCs enter liver cells, a fatty liver appears, which gradually evolves into liver fibrosis and finally induces tumors, leading to liver cancer.

3.1. Epidemiological investigation of MCs

Epidemiological investigations have shown that MCs can cause acute and chronic poisoning in humans by damaging the cardiovascular system, liver, and other organs (Cao et al., 2019). As a liver tumor promoter, MCs strongly inhibit protein phosphatase activity (Shi et al.,

2021).

The WHO has established a tolerable daily intake (TDI) of 1 µg/L for MCs in drinking water (Chorus and Welker, 2021). When the amount of MCs ingested over a short period exceeds a certain threshold, it often leads to acute poisoning and even death. In 1996, 76 people died in Brazil from acute MC poisoning (Carmichael et al., 2001; Shi et al., 2021) caused by contamination. Several healthy boys in Dane, Wisconsin, USA, died after experiencing seizures while swimming in a pool rich in MCs (Weirich and Miller, 2014). A review of these acute poisoning cases reveals that acute exposure to high doses of MCs can cause serious organ damage and even death.

Low-dose and long-term exposure to MCs can lead to chronic liver poisoning symptoms in the human body, gradual morphological changes in the liver, functional damage, and even liver tumors. In Florida, where severe blooms occurred in 2016 and 2018, studies measured MC concentrations in nasal mucosal swabs and found average values as high as 0.77 + 0.88 ppb in residents who lived near waterways and outdoor workers who worked near water for a long time (Schaefer et al., 2020). Children and young people are particularly vulnerable compared to adults because of their lower body weight, risky behavior, and toxic effects of MCs on development. Previous community-wide cyanobacterial toxin outbreaks led to greater morbidity and mortality rates in children than in adults (Weirich and Miller, 2014). MC exposure in the Three Gorges Reservoir area was significantly and positively correlated with changes in serum MC-LR levels and changes in liver function indices (Lv, 2019). Additionally, MC-LR can induce liver cancer. Studies using questionnaires to obtain basic information about lifestyle and disease history revealed a serum MC-LR adjusted liver cancer hazard ratio of 2.9 (95 % confidence interval [CI], 1.5–5.5) in 214 patients, which demonstrated the clear relationship between elevated levels of MC-LR and the risk of hepatocellular carcinoma (Zheng et al., 2017). With the frequent occurrence of cyanobacterial toxin poisoning incidents, scientific research teams worldwide have conducted detailed analyses and investigations on major MC poisoning incidents and in-depth research on the poisoning mechanism, as shown in Table 2.

3.2. MCs trigger metabolic imbalance in the liver

Epidemiological data show that MCs can cause irreversible damage to the liver by disrupting metabolic balance. Among these, glucose and lipid metabolism represent the main functions of the liver and are

extremely sensitive to MCs.

3.2.1. Effect of MCs on glucose metabolism

MCs affect glucose metabolism in the liver. The liver is the most important metabolic organ in humans, plays a pivotal role in the metabolism of substances, and is essential for the regulation of glucose homeostasis. MC-LR interferes with interrenal cortisol production by regulating the expression of genes in the hypothalamic–pituitary–adrenal axis, HPA or HTPA Hypothalamic–Pituitary–Interrenal axis, thereby impairing glucose metabolism in the liver (Wang et al., 2020). A study in which adult male zebrafish were immersed in 25 µg/L MC-LR solution revealed reduced expression of hepatic brain nuclear receptors (GR), presumably due to the potent inhibition of genes downstream of glucokinase. Moreover, the results showed that long-term exposure to MC-LR resulted in lower production of glucose relative to glucose-cleaved sugars in the liver and revealed that heterozygous-related genes, glycogenolysis-related genes, and glycolysis-related genes were suppressed (Wang et al., 2020).

3.2.2. Effect of MCs on lipid metabolism

MC-LR can affect lipid metabolism in the liver and increase liver fat weight. Studies have shown that chronic MC-LR exposure induces the activation of endoplasmic reticulum and sterol regulatory element-binding protein (SREBP), which drives the expression of genes related to hepatic lipid and cholesterol metabolism (e.g., srebf1, srebf2, and fasn), thus leading to increased hepatic adipogenesis and steatosis (Qin et al., 2015). One study showed lipid vacuole accumulation and increased abdominal lipid weight in the liver of mice following gavage with MCs for 28 d. Moreover, serum nuclear magnetic resonance analysis showed that serum metabolite triglycerides, unsaturated fatty acids (UFAs), and very low-density lipoprotein levels were altered after MC-LR treatment, indicating that MCs severely affected hepatic lipid metabolism in mice (Zhang et al., 2016). Chronic low-dose exposure to MCs promotes liver inflammation by inhibiting fatty acid oxidation and hepatic lipoprotein secretion, leading to nonalcoholic steatohepatitis (NASH) (He et al., 2017b). MCs can also be ingested through drinking water and cause damage to the body. A study that provided MC-contaminated drinking water to mice found that high doses of MC-LR poisoning group of mice serum alanine amino transferase (ALT) level and high and low doses of MC-LR poisoning group serum aspartate amino transferase (AST), the glucose level is relatively high, high and

Table 2
Summary of MC poisoning in humans and other organisms.

Survey time	Location	Survey object	Main route of exposure	Survey result	References
1996	Brazil	131 patients at the Caruaru Dialysis Center	Directly injected into the blood through a vein; Drinking water intake	76 people died	(Shi et al., 2021)
2002	Dane, Wisconsin	Several healthy boys	Exposure by contact	Epileptic death	(Weirich and Miller, 2014)
2016	Florida	Outdoor workers near waterways	Exposure by contact	Excessive levels of MCs	(Schaefer et al., 2020)
2018	Florida	Outdoor workers near waterways	Exposure by contact	Excessive levels of MCs	(Schaefer et al., 2020)
/	Three Gorges in China	Nearby crowd	Exposure by contact	Liver function index changed significantly	(Lv, 2019)
2013–2016	Chongqing, China	214 newly diagnosed HCC patients in the hospital	/	Risk ratio of liver cancer to general patients was 2.9	(Zheng et al., 2017)
2007	Yunnan, China	A local resident	Dietary intake	Die of poisoning	(Shi et al., 2021)
1931	United States	Local residents	Exposure by contact; Drinking water intake	A series of diseases are caused in the towns along the river	(Gintert et al., 2019)
/	Zimbabwe	Children in a particular region	Drinking water intake	Annual occurrence of gastroenteritis	(Weirich and Miller, 2014)
2007	East China	Residents who drink water from ditches and ponds	Drinking water intake	Liver cancer mortality rate is >10 times higher than in other regions	(Chuan-hui et al., 1998)
2007	Big Lake and Money Lake	Sediments in the Great and Money Lakes	/	Excessive levels of cyanobacteria toxins in water organisms	(Chen et al., 2008)

(Note: “/” in this table indicates that the section was not redundantly described in the literature, so there is no relevant data description.)

low doses of MC-LR poisoning group serum triglyceride (TG) level is relatively low. Moreover, this study showed that drinking MC-LR-contaminated water can cause disorders of glucolipid metabolism, lipid metabolism, insulin resistance, and inflammatory response in mice (Bi-xia et al., 2011). Changes in the gut microbial community structure and function induced by MC-LR may also contribute to disturbances in lipid metabolism (Qin et al., 2010).

3.2.3. Effect of MCs on iron homeostasis

The liver is the leading storage site for trace iron elements and the body's primary regulator of iron homeostasis; thus, it plays a vital role in metabolic homeostasis. Exposure to 25 µg/kg MCs induced iron accumulation in the liver of C57BL/6 mice with anemia, which was similar to the results observed for thalassemia; moreover, dysregulation of iron homeostasis may be due to MC-induced downregulation of Hamp1 and associated with IL6-STAT3 and BMP-SMAD-mediated signaling pathways (Wu et al., 2018). The MC-LR-induced downregulation of Hamp1 may be partially caused by hypoxia, which induces the downregulation of hepcidin and leads to an imbalance in iron homeostasis.

3.3. MCs induce hepatocyte death

MC-induced cell death usually manifests as programmed cell death or necrosis. MC-LR strongly inhibited protein phosphatase 1 and protein phosphatase 2 A, leading to hyperphosphorylation of the cytoskeleton and apoptosis and ultimately to the disintegration of animal hepatocytes (Bang-jun, 2016). Exposure to 15 µg/kg MC-LR induced apoptosis in hepatocytes via the per-Bax-cyt c-caspase 9-caspase 3 pathway in mitochondria (Li et al., 2015). After exposure to 25 µg/kg MC exposure for two weeks, a large amount of ROS was generated in hepatocytes. ROS generation promoted the expression of apoptotic genes (e.g., p38, JNK, and bcl2), and the rate of apoptosis was significantly increased (Jin-xia, 2019). In one study, intraperitoneal injection of 25 µg/kg MC-LR into mice for 28 d resulted in decreased α -tubulin levels, activation of p38/MAPK and CaMKII pathways in the liver, and ultimately caused apoptosis due to destruction of the cytoskeleton (Bang-jun, 2016). There is considerable evidence that MC exposure induces apoptosis in hepatocytes primarily by activating pathways that disrupt the cytoskeleton, thereby inducing apoptosis and reducing liver function.

MC-LR has been shown to induce programmed cell death and necrotic prolapse (Wu et al., 2019). Necrosis is a passive mode of cell death that is not regulated by gene expression. It manifests as the swelling of organelles, rupture of plasma and nuclear membranes, and release of contents. It is generally considered pro-inflammatory and immunogenic. When cells die, the integrity of the cell membrane is destroyed and its contents are released, causing an inflammatory response. Recent studies have found that the caspase-independent processes of necrosis, extracellular signaling (death receptor-ligand binding), and intracellular cues (foreign microbial nucleic acids) may cause programmed necrosis (necroptosis), which can be significantly inhibited by caspase activity. Hepatocyte necrosis is manifested by organelle swelling, plasma and nuclear membrane rupture, and release of contents and is generally considered to be pro-inflammatory and immunogenic. Necrosis has been defined as a nonprogrammed form of cell death caused by significant chemical or physical damage. Several recent studies have shown that MCs induce cell death through necrotic and apoptotic mechanisms (Shi et al., 2021). ROS are considered a causative factor in necrotizing prolapse and could be a new drug target for treating patients exposed to MCs (Wu et al., 2019). It has been shown that rip3-induced mitochondrial ROS production in the liver leads to necrotic prolapse in response to TNF- α stimulation, and MC-LR-induced ROS overproduction leads to the production of pro-oxidants SOD1, MAOA, and NOX4, resulting in the dysregulation of GPX1 (Wu et al., 2019). Thus, ROS appear to be the driving force underlying cell necrosis. Under the action of MC-LR, ROS are produced in large quantities, and the oxidants SOD1, MAOA, and NOX4 and antioxidant GPX1 are misregulated.

Therefore, the knockdown of MAOA, NOX4, and SOD1 may be a key therapy for reducing MC-LR-induced hepatocyte necrosis.

3.4. MCs induce nonalcoholic steatohepatitis

NASH causes extracellular matrix remodeling in the liver and is a risk factor for hepatocellular carcinoma (HCC). A previous study found that mice appeared to have signs of NASH when a MCs diet was induced (Arman et al., 2019). One study demonstrated a surge in liver fat content ($15.3 \pm 1.6\%$) and significant effects on the TBARS, SOD activity, and glutathione content in the liver of mice that orally ingested toxic MCs (Sedan et al., 2015). Circulating alkaline phosphatase levels and levels of the liver-damaging enzymes ALT and alkaline phosphatase (ALP) were significantly elevated in the nonalcoholic fatty liver disease (NAFLD) mouse model (Leprdb/J) compared to healthy mice, even early death occurs (Lad et al., 2019). Healthy and NASH liver tissues have similar susceptibilities to MC toxicity. The results show that MCs cause dysregulation of crucial signaling pathways (PI3K/AKT/NF- κ B pathway) and dysregulation of the phosphorylated proteome in the liver of NAFLD mice (Lad et al., 2019). In individuals with NAFLD, MCs' metabolic and detoxification functions of MCs are significantly impaired, ultimately leading to irreversible outcomes (Lad et al., 2022). Chronic MC-LR exposure in preexisting diet-induced NASH can produce a more severe NASH phenotype, such as reduced steatosis, increased inflammation, and increased fibrosis (Arman et al., 2019). MCs trigger an inflammatory response network mediated by immune cells in many organs by inhibiting protein phosphatase-1 (PP1) and -2A (PP2A), producing unusual cellular protein phosphorylation that may activate protein kinase c, which in turn activates phospholipase A2 and cycloperoxidase and triggers inflammation (Carvalho et al., 2010). When treating liver inflammation caused by MCs, inflammation may be reduced by inhibiting protease c. This can be used as a new drug target to provide a new idea for the subsequent treatment of liver inflammation caused by MCs.

3.5. MCs induce liver fibrosis

Among MC-induced diseases, liver-related diseases have the greatest impact and are the most widespread. However, most chronic liver diseases, including hepatocellular carcinoma and intrahepatic bile duct carcinoma, have been shown to be associated with liver fibrosis (Yan et al., 2020). Exposure to MCs leads to the activation of hepatic stellate cells (HSCs) in the liver tissue into fibroblasts, leading to liver fibrosis (Fig. 2) (Gu et al., 2020). However, it has also been shown that chronic MC exposure is unlikely to cause chronic liver disease in the presence of a normal liver or to exacerbate existing liver disease in the presence of persistent hepatitis (Labine et al., 2017). MCs induce liver fibrosis through the activation of HSC, which is closely related to the Hh signaling pathway (Zhang et al., 2017). In mouse experiments, chronic MC exposure caused an epithelial-mesenchymal transition in mouse lung epithelial cells (Wang et al., 2016). Thus, MCs may destroy the liver hepatic cords and induce apoptosis, leading to liver fibrosis with various pathological changes leading to liver scarring (Lee et al., 2010). Exposure to MCs activates stellate cells in the liver tissue. Activated stellate cells turn into myofibroblasts and the considerable accumulation of myofibroblasts eventually leads to liver fibrosis.

It was found that the Hedgehog signaling pathway transcription factor Gli2 plays a decisive role in liver fibrosis, and that Gli2 can be considered a drug target to alleviate MC-induced liver fibrosis, providing a new idea for the development of drugs to alleviate MC-induced liver fibrosis.

3.6. MC exposure is a risk factor for liver cancer

MCs act as strong promoters of liver tumors and potentially promote tumorigenesis and progression. Under certain conditions, MCs induce hepatocarcinogenesis by activating the liver oncogenes. The liver has a

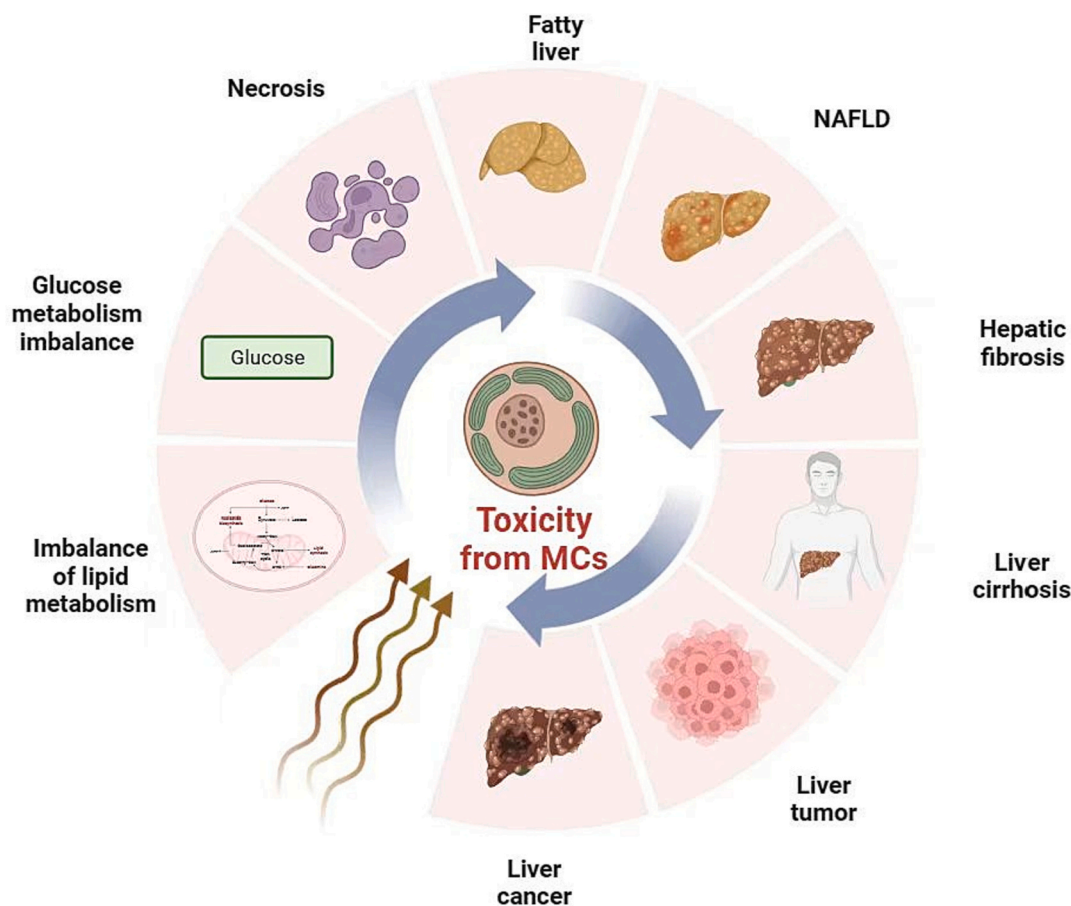


Fig. 2. Mechanisms of liver toxicity from MC exposure. The entry of MCs into cells first causes metabolic disorders of the liver, subsequently changes the cell morphology, and induces apoptosis. With a large number of MCs entering the cells, a fatty liver is observed, which gradually evolves into liver fibrosis and finally induces tumors, leading to liver cancer.

specific regenerative capacity. When this limit is exceeded, local inflammatory liver tissue damage, necrosis, and liquefaction can occur. To a certain extent, it causes hepatitis, the most common of which is hepatitis B infection. When hepatitis B infection, hepatocyte degeneration, and necrosis occur, the normal liver tissue “collapses”. Subsequently, the body’s regenerative function will regenerate some fibers to fill the “collapsed” parts so that the necrotic areas heal. However, if the hepatocytes continue to necrotize and the fibers in the liver continue to regenerate, the regenerated fibers replace most of the liver tissue, and they do not function as normal liver cells; the liver becomes hard and small, forming cirrhosis, which eventually leads to cancer.

3.6.1. MCs cause an inflammatory response in the liver

Oxidative stress is one of the significant toxic manifestations of MCs, which induces cytotoxic damage either directly or indirectly through the induction of apoptosis and the release of pro-inflammatory cytokines. Exposure to MCs induced the expression of the pro-inflammatory factors IL-6, IL-18, and MIF and the anti-inflammatory factors IL-4 and IL-10. It also inhibited the expression of the anti-inflammatory factor IL-2 in mouse liver, which affected the pro-inflammatory/anti-inflammatory balance in the liver and eventually led to liver inflammation and impaired liver function (Bang-jun, 2016). After intraperitoneal administration of 10 $\mu\text{L/kg}$ MC-LR to rats for 14 consecutive days, biomarkers (aspartate transferase, alanine transferase, and alkaline phosphatase) were reduced in the rat serum and inflammatory responses were observed in the rat liver (AlKahtane et al., 2020). MCs were shown to upregulate PfHMGBl protein in the liver of juvenile eels, which was followed by hepatocyte disintegration, nuclear atrophy, mitochondrial

and rough endoplasmic reticulum disruption, and liver injury (Fig. 2) (Wang et al., 2021).

3.6.2. Induction of tumorigenesis by MCs

MCs are major cancer promoters in the natural environment, and studies have shown that MC-LR at subacute staining concentrations (0.01, 0.05, 0.5, 10, and 50 μM) promotes the expression of the oncogenes c-fos, c-jun, c-myc, c-met, and N-ras in HepG2 cells at the transcriptional and protein levels while suppressing the expression of the oncogene PTEN (Li et al., 2017). MC-LR is a novel liver tumor promoter that mediates tumorigenesis by inhibiting protein phosphatase type 1 and 2A activity, and a previous study showed that the tumor incidence increased in rats exposed to MC-LR, with this response triggered by N-nitrosodiethylamine (Rie Nishiwaki-Matsushima and Ohta, 1992). Nuclear factor erythroid 2-related factor 2 (Nrf2) plays a crucial role in regulating the protective antioxidant response of the body to various environmental toxicants, and it has been suggested that MC-LR can induce Nrf2 activation in hepatocellular carcinoma cells and promote cancer cell growth (Gan et al., 2010). Future studies should investigate Nrf2 as a drug target to provide insights for cancer treatment.

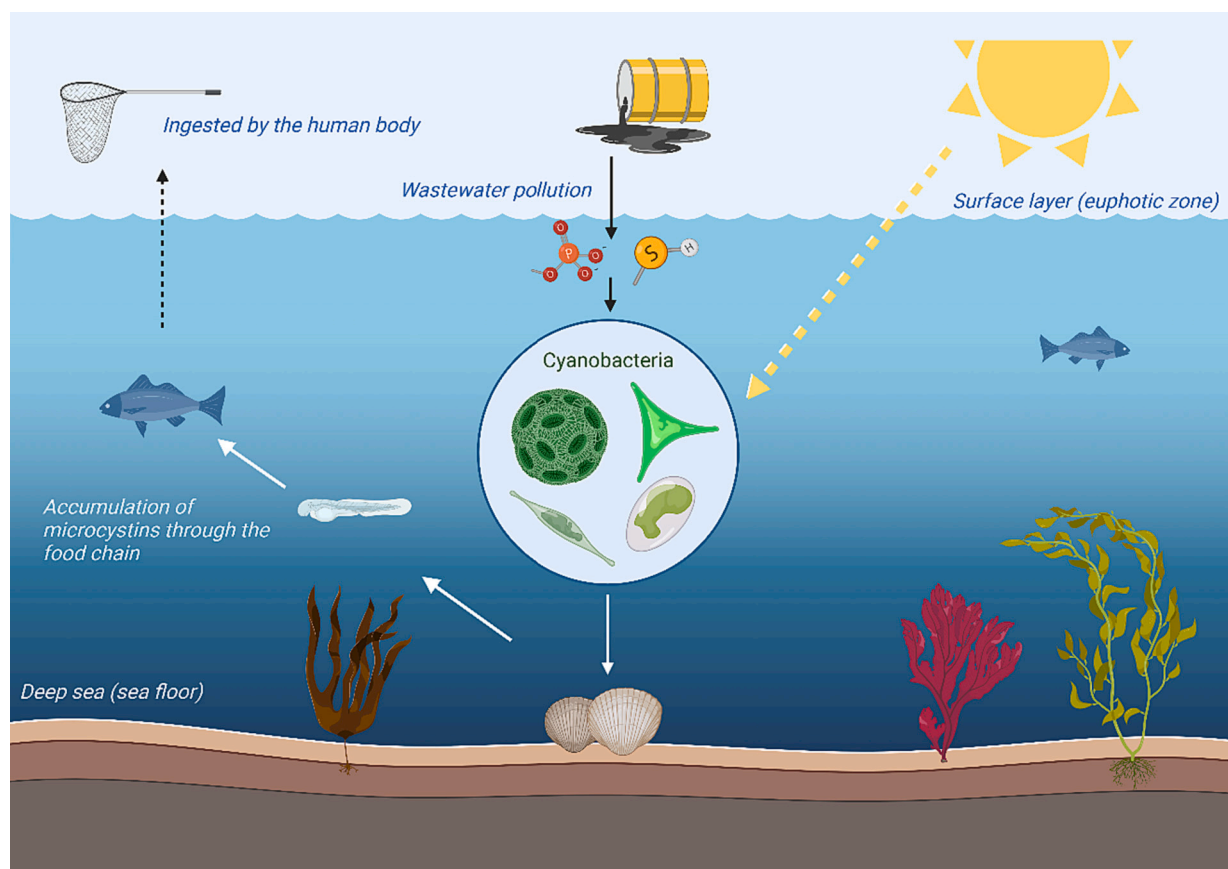
As a metabolic center, the liver is greatly affected by exposure to MCs. When MCs enter liver cells in the early stages, cell metabolism becomes chaotic and apoptosis is induced. As a large number of MCs enter the cells, the liver becomes overloaded and a fatty liver gradually appears. A large amount of fat accumulation in the liver is often accompanied by liver stellate cell hyperplasia, glycoproteins, activated collagen, and other extracellular matrix components as well as excessive deposition in the liver tissue, that is, the formation of fatty liver fibrosis.

4. Toxic effects of MC exposure on hepatocytes

4.1. Enrichment characteristics of MCs in hepatocytes

Similar to fish, some freshwater benthic animals (e.g., mussels, clams, snails, and shrimp) also show high tolerance to MCs, and they

present a strong enrichment effect for toxins in the body. Therefore, MCs may exert toxic effects on juvenile mussels at environmentally relevant concentrations. The environmental exposure distribution was created using Canadian and Canadian-US Great Lakes MC measurements, and a median lethal concentration (LC50 value) of 2.1 µg/L was found in mussel larvae after 28 d of exposure (Gene et al., 2019). The eastern oyster (*Crassostrea virginica*) was not able to depurate MCs over 3–4 d of exposure compared to the Asian clam (*Corbicula flumina*), and an order of magnitude more MCs were accumulated in the oyster compared with that in the clam (Straquadrine et al., 2022). Thus, Asian clams are superior to eastern oysters in eliminating MC accumulation in vivo, and oysters are likely vectors of human hepatotoxic shellfish poisoning because they easily accumulate MCs. Cyanobacterial blooms strongly affect freshwater zooplankton communities. MC exposure was found to significantly increase the abundance of proteins related to reproduction, development, superoxide radical scavenging, and motor activity and cause a large number of differentially expressed proteins, which were mainly related to fat, carbohydrate, amino acid, and energy metabolism (Lyu et al., 2016; Shahmohamadloo et al., 2020b). MC-LR showed a decreasing trend in the hepatopancreas (1.06–7.42 mg/g dry weight), digestive tract (0.8–4.54 mg/g dry weight), gonads (0–2.62 mg/g dry weight) and gastropods (0–0.06 mg/g dry weight) of freshwater snails (Chen and Xie, 2005). In a study on copper-rusted ringed snails from Chaohu Lake, the accumulation of MCs in the hepatopancreas (mean value of 4.14 µg/g dry weight) was much greater than the enrichment in the gonads (mean value of 4.14 µg/g dry weight). Studies have assessed the risk of MC enrichment in invertebrates by measuring the survival, growth, and reproduction of pelagic and benthic organisms exposed to MC-bound cells (Shahmohamadloo et al., 2020a). Such research efforts produce value information, and the generation of a refined complete



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assessment system could help assess the direct risk posed by cyanobacterial blooms to invertebrates as well as the indirect risk posed to consumers who depend on algae as an important part of their diet.

4.2. Effects of MC exposure on hepatocyte structure, appearance, and function

MC exposure leads to the following changes in hepatocyte ultrastructure: wrinkling of the rough endoplasmic reticulum, secondary expansion of mitochondria, cytoplasmic vacuole-like changes, and refolding of the plasma membrane (Wen-guo and Qing-guo, 2014). Exposure to MCs can lead to cellular DNA damage and gene mutations, and it can also induce the development and metastasis of hepatocellular carcinoma; thus, it poses a serious health risk (Yin-tao, 2021). Animal testing has shown that exposure to low-dose MCs can cause malignant transformations in the animal liver and some degree of liver and tumor damage. Moreover, long-term MC exposure can induce the malignant transformation of normal human hepatocytes, which is primarily caused by the elevated expression of gankyrin induced by MCs (Chen et al., 2013). Comet and 8-OHdG level assays showed that MC-LR induced DNA double-strand breaks in hepatocytes and increased the range of 8-OHdG level change from 1.02 ± 0.03 -fold (2.5 nmol/L group, $P < 0.05$) to 1.51 ± 0.07 (10 nmol/L group, $P < 0.05$), which led to DNA damage (He et al., 2021). The accumulation of large amounts of MCs eventually causes irreversible damage to the morphology and function of liver cells and destroys the genetic material and ultrastructure of the cells. Exposure to MCs also affects the homeostasis of cytoskeletal gene expression and may lead to dysfunctional cytoskeletal assembly, which is closely associated with altered cell morphology (Chen et al., 2013).

The endoplasmic reticulum is considered one of the factors that can mediate apoptosis. Studies have found that CHOP mRNA expression in cells significantly increases following MC-LR exposure (Qin et al., 2010). CHOP is involved in ERS-induced apoptosis. CHOP is a specific marker of the ERS response and represents the first pro-apoptotic transcription factor to be identified during the activation of the ERS signaling pathway. CHOP may induce apoptosis through the ERS-UPR signaling pathway. Blocking of this pathway could be a solution to rescue apoptosis and exert therapeutic effect.

4.3. MCs affect intrahepatocellular signaling pathways

MC-LR can induce excessive proliferation of hepatocytes by inhibiting the activation of the Akt/mTORC1/S6K1 pathway via PP2A activity. The Akt/ β -catenin and ERK/JNK/p38 MAPK pathways may also be involved in this process. Activation of the Akt/mTORC1/S6K1 pathway in HL7702 cells was initiated by exposure to MC-LR for 1 h, in which the activation of upstream Akt and inhibition of PP2A activity were involved in the activation of S6K1 by MC-LR (Jing-hui, 2016). MCs can induce damage to the mitochondrial structure and function in hepatocytes through the interleukin IL-6-STAT3, BMP-SMAD, and ERS pathways (Fig. 5) (Jin-xia, 2019; Qin et al., 2010). Interleukin-6 intracellular signaling first activates Janus kinase (JAK), which initiates an enzymatic reaction. The FERM-domain of JH7-JH5 mediates its interaction with gp130 in the N-terminal homologous domain (JH) of JAK. The binding of IL-6 to its receptor is followed by the activation of JAK by auto-phosphorylation, mainly at the double tyrosine motif of the activation loop within the C-terminal JH1-structural domain of the kinase. IL-6 induces Hck and Lyn expression, interacts with gp130, and subsequently activates Src kinases (Hck, Lyn, and Fyn). Activated JAK kinase phosphorylates the tyrosine motifs within the cytoplasmic fraction of gp130. Signal transduction and transcriptional activators STAT3 and STAT1 are recruited to gp130. Increased IL-6 levels lead to STAT3 activation and decreased MAPK activation. In the classical MAPK pathway, MC exposure induces p38 and MAPK activation, thus leading to JNK and ERK phosphorylation, impaired cell assembly, and altered cell morphology. The role of MAPK in regulating MC-induced hepatic

cytoskeletal rearrangement and polymerization was demonstrated by the finding that MC-induced microtubule polymerization was inhibited by p38 MAPK inhibitors (Fig. 4).

4.4. Interaction of MCs with protein phosphatases PP1 and PP2A in hepatocytes

PP1 and PP2A are the main target proteins of MCs, and after binding to MCs and their analogs, their activity is inhibited, leading to toxic effects. PP1 and PP2A belong to the PPP family and contribute to most of the serine/threonine activity in hepatocytes (Liu and Sun, 2015). PP2A is an essential serine/threonine phosphatase that is vital for the regulation of hepatocyte division and gene expression. The activities of serine/threonine-specific protein phosphatases (PSTPs) PP1 and PP2A are inhibited by the uptake of MCs (Fontanillo and Köhn, 2018). The inhibition of PP1 and PP2A activity induces protein dephosphorylation; however, after the protein is dephosphorylated, the activity of PP1 and PP2A cannot be restored. Subsequently, MCs alter hepatocyte metabolism, leading to MC toxicity in hepatocytes as indicated by inflammatory and irritant reactions and liver-related diseases. In addition, MCs can induce the production of ceramide, an essential intracellular secondary messenger that regulates various cellular mechanisms, including PP2A activation and cytoskeletal destabilization. It has been found that ceramide may mediate the MC-LR-induced upregulation of PP2A activity and PP2A regulatory subunit protein levels in HEK293 cells (Li et al., 2012). This demonstrates the importance of PP1 and PP2A in MC-mediated acute intoxication and the importance of identifying PP substrates to thoroughly characterize the MC pathway.

4.5. Hepatocyte oxidative damage due to MC exposure

Oxidative stress is the primary biochemical feature of MC toxicity, and it gradually causes oxidative damage to organisms with increasing exposure time and dose. Oxidative damage is characterized by a redox imbalance between free radical levels and cellular antioxidant capacity during cellular oxidative stress, a relative lack of cellular antioxidant capacity, and an increase in strongly toxic reactive oxygen radicals that damage biological macromolecules, causing cellular and tissue damage. After 7.5 and 15 $\mu\text{g/kg}$ MC-LR exposure for 21 d, ROS were significantly elevated and caused oxidative damage to the mouse organism (Bang-jun, 2016). MCs can cause acute oxidative damage in the liver tissues of mammals and fish, accompanied by a significant increase in lipid peroxidation levels (LPO) and antioxidant enzyme (e.g., SOD, CAT, GPx, and GR) activity were significantly altered (Yan-yan, 2009). As the concentration of MCs gradually increases, the level of intracellular ROS in the tilapia liver gradually increases, resulting in the activation of SOD and CAT (Jiazhang et al., 2011). Alcohol contamination by high levels of MCs can lead to hepatic oxidative stress and induce liver-related diseases (Xiao-feng et al., 2007). In the hepatopancreas of *Litopenaeus vannamei*, exposure to MCs resulted in a gradual increase in the rate of lipid peroxidation, which inhibited the activity of antioxidant-related enzymes, resulting in oxidative stress and severe oxidative damage to the hepatopancreas within 72 h (Lan-ying et al., 2022). In one study, it was reported that in rat liver cells, exposure to MCs initially caused a surge in mitochondrial Ca^{2+} levels, which initiated a permeability transition of the mitochondrial outer membrane and enhanced ROS production (AlKahtane et al., 2020). Acute exposure to high doses of MCs results in liver injury and elevated serum levels of liver markers that originate from oxidative stress in hepatocytes because of the reduction in antioxidant enzyme activity caused by ROS generation.

Reports have shown that 2.5 nmol/L MCs significantly reduced ATP levels in mouse hepatocytes (by approximately 22 %), whereas increasing the MC concentrations decreased ATP production. The subsequent mitochondrial DNA damage led to the upregulation of p53 levels, which induced mitochondrial dysfunction (He et al., 2021). After treating mouse hepatocytes with 50 $\mu\text{g/kg}$ b.w MCs, the expression of

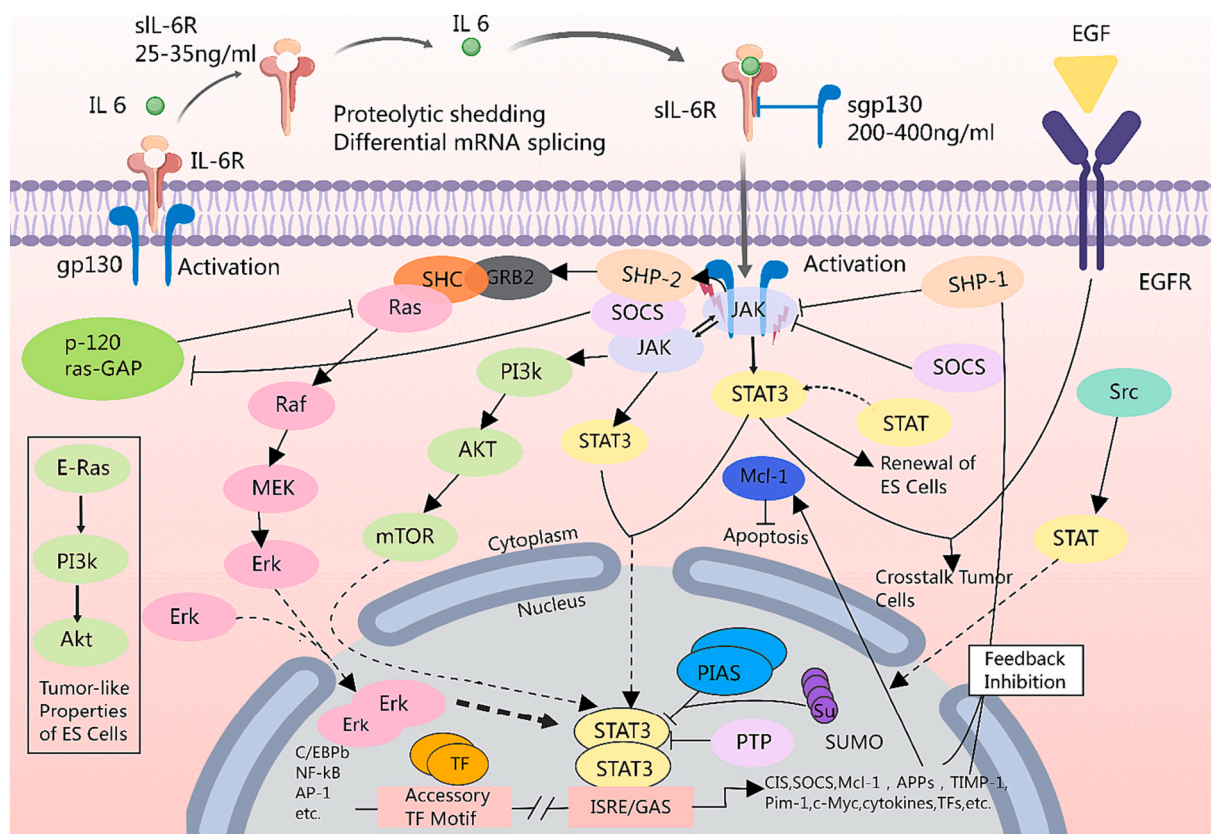


Fig. 4. MCs contribute to hepatocyte cycle disruption and carcinogenesis by activating the IL-6/JAK/STAT pathway.

the DNA ladder and Bcl-2 protein levels were upregulated, the mitochondrial membrane potential of liver tissue decreased, and the ROS level increased (Jing et al., 2011). High concentrations of MCs activate numerous caspases, induce apoptosis, and cause liver inflammation. The mitochondrial ultrastructure was altered and lipid peroxidation damage was observed after intraperitoneal injection of 12.5 and 50 µg/kg body weight doses of crude extracts of MCs to rabbits, and the toxic effects caused by the high concentrations of MCs were irreversible (Yan-yan, 2009). After intraperitoneal injection of MCs, the endoplasmic reticulum was lysed in mouse hepatocytes, ribosome degranulation occurred, and deformation and fragmentation of the mitochondrial cristae was observed (Ming bo et al., 2005). Long-term MC exposure will alter mitochondrial DNA content and integrity and increase 8-OHdG levels in hepatocyte DNA, thereby disrupting the integrity of mitochondrial DNA and nuclear DNA (NDNA), altering mitochondrial and nuclear gene expression, and impairing mitochondrial and hepatocyte functions (Li et al., 2015). MCs disrupt the energy and related metabolism of the mitochondrial respiratory chain and phosphorylation system and induce apoptosis by producing large amounts of ROS and oxidative damage. All eight mitochondrial genes associated with oxidative phosphorylation (OXPHOS) were significantly upregulated in MC-exposed mice, and ROS levels were significantly elevated, which caused mitochondrial swelling and DNA damage (Fig. 5).

5. Toxic effects of the combined exposure of MCs and other harmful substances on the liver

In nature, MCs coexist with various toxic chemical substances. With the rapid development of science and technology, many toxic pollutants have entered water sources and acted synergistically with MCs. Such combinations pose a serious threat to numerous organisms. Exposure to MCs along with other toxic compounds causes structural and functional damage to the liver and ultimately endangers human health. One study

found that the markers of liver injury glutamic oxaloacetic transaminase (AST) and glutamic alanine transaminase (ALT) were significantly higher under the combination of MCs and aflatoxin compared with that observed with MC and aflatoxin exposure alone, and the OR values were also continuously higher (Liu et al., 2017). Mice co-exposed to low-dose MCs (0.0024 mg/kg) and aflatoxin B121d (under subchronic exposure) showed a synergistic effect with a slightly higher overall expression of five biochemical parameters in the serum and liver than the single-chemical group (Yun et al., 2021). A study that performed acute and chronic toxicity experiments on the zooplankton *Daphnia magna* to determine the effects of single and combined exposure to polycyclic aromatic hydrocarbons (Phe) and MCs found that the toxicity of Phe to *Daphnia magna* was much higher than that of MC-LR and the acute combined toxicity of both pollutants showed synergistic effects at high concentrations (Xiang, 2020). In addition to the MCs produced by cyanobacteria, the dramatic increase in nitrite levels in the water column owing to anoxic conditions is often overlooked during cyanobacterial blooms. However, MCs and nitrites can coexist in water and potentially have combined toxic effects on aquatic organisms. A previous study exposed adult male zebrafish to different combinations of MC-LR (0, 3, and 30 µg/L) and nitrite (0, 2, 20 mg/L) solutions for 30 d. The results showed that the decreased dopamine content, acetylcholinesterase activity, and mRNA transcript levels of related genes had synergistic neurotoxic effects on the structure and neurotransmitter system of the fish brain and that the disruption of antioxidant capacity caused by these two factors could be a potential synergistic mechanism (Li-ping, 2021). In industrially contaminated water bodies, the disinfection by-products of nitrosamines (NAs) in drinking water after filtration and disinfection have become a significant source of contamination. EC109 cells were exposed to MCs and NAs in combination, and the results suggested that this combination was highly carcinogenic to the body and may promote the malignant transformation of cells by inhibiting the expression of PP2A and promoting the expression of Ras to activate

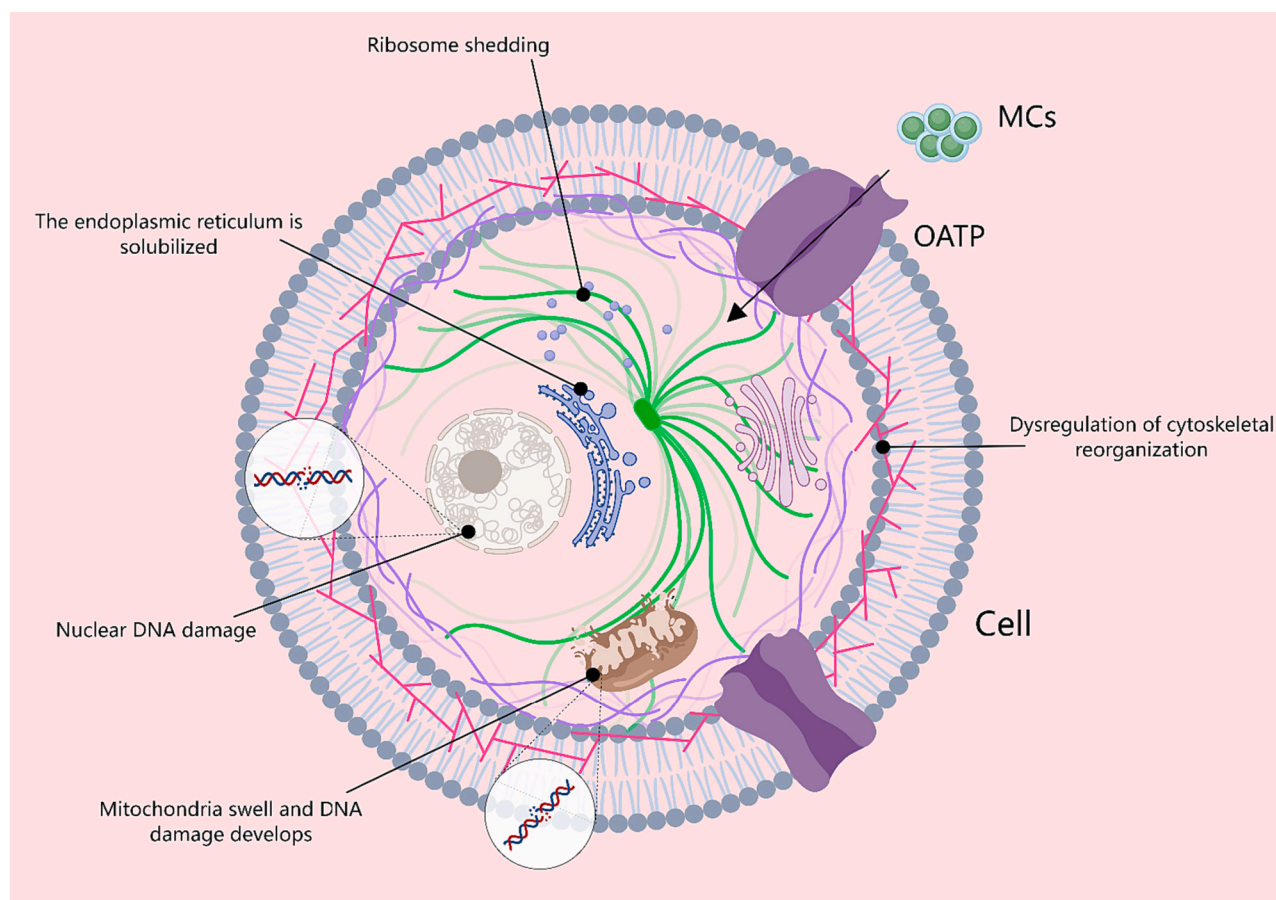


Fig. 5. Schematic diagram of the sub-microstructure of MC-damaged cells. MCs enter cells through OATP proteins, thereby causing varying degrees of damage to the cytoskeleton, nucleic acids, endoplasmic reticulum, and mitochondria.

downstream signal transduction pathways (Fei-fei, 2016). In a study that applied a combination of three typical contaminants, namely, cadmium (Cd), MC-LR, and diclofenac (DCF), to crayfish, severe oxidative stress due to toxic contaminants was observed in the primary target organs of the crayfish, including the hepatopancreas, gills, and gut; moreover, pathological changes were observed in all tissues of the hepatopancreas, including luminal swelling and vacuolar structures (Yu, 2020). In zebrafish hepatocytes, chronic 90 d MC-LR exposure resulted in edema. Nanoplastics (NPs) further aggravate hepatocyte edema and exacerbated MC-LR-induced overexpression of related genes in the Nrf2 pathway, which led to increased oxidative stress in zebrafish liver, thereby aggravating liver injury (Xiao-dong, 2021).

In conclusion, current population growth and increasing industrial modernization have resulted in the discharge of many toxic pollutants into the environment, causing the generation of many harmful substances. Recent studies have shown that MCs and other pollutants have a combined negative effect on the liver. Research on the effects of toxins combined with other pollutants in the environment can better reveal the ecotoxicity of toxic pollutant compounds in environmental media and provide a theoretical basis for the prevention and control of global ecological and environmental problems.

6. Analytical methods to diagnose MC toxicity to the liver

Current experimental methods for investigating the toxic effects and mechanisms of MC exposure in the liver include animal stress experiments, ecotoxicological experiments, and epidemiological investigations. In animal stress experiments, MCs are usually administered to experimental animals (mice, rats, and zebrafish) (Gu et al., 2020; Qin

et al., 2010; Shuai et al., 2017; Wang et al., 2010a) oral ingestion (Huang et al., 2011), gavage, and intraperitoneal injections to observe changes in liver tissue structure and function. When conducting epidemiological investigations, the relationship between MC exposure and liver lesions in the population is comprehensively analyzed using observational and experimental methods (Feng et al., 2022; Vareli et al., 2013). Parameter settings such as the exposure dose, route, and duration vary with the type of experimental animal and are also relevant to acute and chronic toxicity studies. Animal stress experiments have the advantages of easy control of exposure conditions, the ability to measure multiple effects, and the ability to explore the role and mechanisms of host characteristics. However, problems are observed, such as individual variability, long cycle time, and cumbersome operation. Cellular experiments have short experimental cycles, allow for the direct observation of cell growth, and permit the control of cell growth status; however, the drawback to such experiments is that the cell culture environment differs significantly from the internal environment of the organism. Moreover, cell experimental results cannot be used to interpret the mechanism of MC toxicity to the liver. The advantages of epidemiological studies are that they are natural exposure experiments, interactions between chemicals occur under actual conditions, and full human susceptibility can be identified. However, the main disadvantages are the high monetary and time costs, they are primarily retrospective studies, exposure is difficult to determine, problems with confounding exposure occur, and only coarse measurement indicators (morbidity or mortality) are available. In conclusion, the combination of three experimental methods, namely animal stress experiments, cell staining experiments, and epidemiological investigations, can more systematically and scientifically explain the effects and mechanisms of

liver toxicity in MCs.

7. Summary and outlook

The toxicity of MC exposure to the liver has been extensively studied. Almost all organisms have been exposed to MCs and their associated toxicity. At the cellular level, various intracellular proteins, molecules, and pathways interact to form a complex and ordered network of toxic signals from MCs, thereby inducing an inflammatory response and causing oxidative damage. Current research mainly focuses on the MC isomer MC-LR despite the abundance of other isomers in natural environments. The toxicity mechanisms and effects of these isomers may be similar or different, which deserves further study and will provide important scientific data. In the substance metabolism pathway, MCs can disturb hepatic lipid and glucose metabolism, thereby aggravating liver damage. The mechanism of hepatotoxicity caused by MC exposure, such as oxidative damage and the induction of apoptosis, varies among species in spite of other commonalities. At the tissue level, MC exposure will directly lead to morphological changes and functional damage in the organism, which directly manifests as liver enlargement, bruising, and increased specific gravity of the liver body. These changes further result in the development of liver fibrosis, nonalcoholic fatty liver, cirrhosis, and liver cancer. At the cellular level, MCs directly or indirectly activate Akt/mTORC1/S6K1, Akt/ β -catenin, ERK/JNK/p38 MAPK, IL-6-STAT3, and other cellular pathways, leading to oxidative damage and sub-micro organelle damage. This damage results in the inactivation of two crucial cell proliferation-related phosphatases, PP1 and PP2A, and trigger malignant proliferation. With increasing industrialization, the combined toxic effects of MCs with other pollutants have been observed, causing irreversible environmental harm.

Numerous animal and cellular toxicity tests and population epidemiology studies have confirmed that MCs pose health risks to humans. However, standardized methods have not been developed to study the toxicity of cyanobacterial toxins in humans, and precise criteria are not available for the threshold of poisoning by MCs in nature (Studer et al., 2022). Cyanobacterial toxins in water bodies can enter humans through the food chain and subsequently cause liver toxicity. Currently, food hygiene standards for cyanobacterial toxins in aquatic and agricultural products are lacking. Though current toxicological research ideas and methods are relatively straightforward, there is a gap in molecular research. Studies on exposure to MCs in the population are mainly focused on acute poisoning. In metabolomics, the role of metabolic disorders in MC hepatotoxicity has not been clarified, and the application of histological methods should be enhanced. Preliminary studies have shown that the combination of MCs with other toxic substances has synergistic effects on the liver. However, few studies have demonstrated these effects. Although many opportunities are available in this research field, the mechanisms and effects underlying the combination of MCs and other toxic substances remain unstudied. The utility of bioinformatics and big data science as emerging technologies and tools are infrequently applied to MC exposure toxicology experiments. Subsequent studies could use bioinformatics for sequence comparisons and model construction to provide insights for MC toxicology research and drug designing. In combined-exposure toxicology studies, more detailed research is required on the molecular mechanisms underlying liver function damage. Moreover, therapeutic methods have not been established for liver damage caused by MC exposure. New drugs or methods must be developed to treat acute MC poisoning at the animal or cellular level and provide systematic criteria for judging MC poisoning in nature.

In summary, MCs represent a global environmental problem. With massive population growth and modern industrial progress, many pollutants have been discharged into the environment. Massive production of toxic substances and eutrophication of water bodies have irreversibly affected the social ecology. This paper reviewed the mechanisms underlying the hepatotoxicity of MCs and identified that few methods are available to alleviate acute MC poisoning, calling for immediate actions.

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CRediT authorship contribution statement

Tong Li: Conceptualization, Software, Writing – review & editing. **Xinting Fan:** Methodology, Data curation, Validation. **Meihan Cai:** Methodology, Data curation, Validation. **Yuanyuan Jiang:** Methodology, Data curation, Validation. **Yaqi Wang:** Methodology, Data curation. **Peishuang He:** Data curation. **Juan Ni:** Data curation, Software, Investigation. **Aili Mo:** Data curation. **Cuiying Peng:** Supervision, Writing – review & editing. **Jun Liu:** Supervision, Conceptualization, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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